

Versa: Keeping financial investments simple and accessible with Google Kubernetes Engine

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About Versa

Versa leverages microservices on its digital cash management platform using Cloud SQL data to understand customer behavior and support workplace

Founded in 2020, Versa helps individual investors access money market funds on its digital cash management platform to maximize the potential of idle cash. Versa gives customers the flexibility to deposit and withdraw their money with earned interest at any time, without sales fees. Licensed by Securities Commission Malaysia as a Registered Market Operator, Versa works with trusted partner Affin Hwang Asset Management to manage funds.

Industries: Financial Services & Insurance

collaboration with Drive and Google

Location: Malaysia

SQL (<https://cloud.google.com/sql/>), Cloud

Google Cloud results

- Uncovers insights from 140 GB+ of data in BigQuery in seconds
- Seamless transition to remote working during COVID-19 pandemic with secure access to Google Cloud resources
- Improves business decision-making with data visualization

Scales microservices quickly better investment experience



About MatrixC

MatrixC has been a Google Cloud and Workspace Specialist partner since 2008. The company provides design solutions, support, and training to help customers realize the business value of Google Cloud. MatrixC supports Malaysia's goal to build a digital economy.

, Sneels

(<https://workspace.google.com/products/sheets/>)
Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)
(<https://marketingplatform.google.com/about/>),

Cloud SQL (<https://cloud.google.com/sql/>)
(<https://marketingplatform.google.com/about/d>

Cloud IAM (<https://cloud.google.com/iam/>)
Google Analytics

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Firebase (<https://firebase.google.com/>)

Firebase Authentication
(<https://firebase.google.com/products/auth>)

Google Workspace
(<https://workspace.google.com/>)

Google Drive
(<https://workspace.google.com/products/drive/>)

on Looker Studio

Whether it's a mom-and-pop inv
retirement nest egg or a millenn
their first car, there has been a ri
globally. In Malaysia, Securities
Malaysia noted an "exceptional
(<https://www.sc.com.my/api/docum?id=3d991fec-887a-4575-a79d-d2ed>
by retail investors in the domest
2020.

Gmail

(<https://workspace.google.com/products/gmail/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Google Marketing Platform

(<https://marketingplatform.google.com/about>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Google Analytics

(<https://marketingplatform.google.com/about/resources/analytics-product-overview/>)

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—Nelson Wong, Chief
Technology Officer, Versa

Versa (<https://versa.com.my/>), a digital cash management platform, makes investing for Malaysians simple by allowing them to open an account, earn interest, and withdraw funds on demand. The company partners with Affin Hwang Asset Management, one of Malaysia's largest asset management firms that manages more than RM80 billion in assets

(<https://affinhwangam.com/who-we-are/overview>).

Through the partnership, Versa enables retail investors to access money market funds, an alternative to fixed deposits, without sales charges or lock-in period.

“Trust is important for building strong relationships. Versa customers expect to see an accurate account balance and make transactions at any given time,” says Nelson Wong, Chief Technology Officer at

Versa. “We designed our digital cash management platform on Google Cloud (<https://cloud.google.com/>) because it is highly available and secure, so customers feel safe using our services.”

Versa runs its containerized application on Google Kubernetes Engine (/kubernetes-engine) (GKE) with more than 20 microservices for different domains such as user services for account-related requests and funds that are required to create and redeem transactions. The company uses Cloud SQL (/sql) and BigQuery (/bigquery) to improve user experience and app performance with smart analytics.

Google Cloud Partner MatrixC (<https://www.matrixc.com>) supported Versa with the configuration and setup of the infrastructure. During the design process, the team brainstormed ideas to answer pressing questions such as, “How do we scale as more customers sign up? How do we secure our network with Google Cloud tools? How do we minimize operating costs?” The end result is a secure, reliable, and scalable environment on Google Cloud launched in January 2021.

“MatrixC saved us countless hours of experimenting to build our platform on Google Cloud,” says Wong. “With the help of MatrixC and Google Cloud, we met the stringent security and operational reliability requirements to obtain a license from Securities Commission Malaysia.”

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Boosts conversion rates with a frictionless user experience on Google Kubernetes Engine

Traditionally, onboarding for financial services is a time-consuming process involving a lot of paperwork for the customer and fund manager. Versa connects microservices through APIs on GKE to automate manual processes and lower the cost of acquisition.

For digital onboarding, the customer registers with a mobile number that Versa verifies with the mobile operator. Versa verifies the customer's identity via an electronic know-your-customer (eKYC) provider that compares the customer's identity document with their selfie for authenticity. Using Optical Character Recognition API, Versa auto-fills the application form with identity data to save the customer's time.

To complete the onboarding process, Versa sends the account opening data to Affin Hwang Asset Management via API to match their required format. By simplifying the onboarding process, customers can set up their Versa account and start investing in minutes.

"We build microservices and APIs on Google Kubernetes Engine for better control over performance. For example, if we experience a spike in customer onboarding, we can easily scale our cloud resources to meet the demand for that service for a smooth user experience," says Wong. "The fully managed Kubernetes service frees up developers to

improve the product, instead of spending time on provisioning and managing the infrastructure.”

“Looker Studio makes it easy for us to turn data into graphs without coding, so developers can focus on improving the platform instead of creating reports from scratch. From operations to customer service, everyone at Versa uses Looker Studio to understand how the company is performing over time and work

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—Nelson Wong, Chief
Technology Officer, Versa

Improving data-driven decisions with Cloud SQL and Looker Studio

For a complete view of the customer journey, from onboarding to making deposits, Versa utilizes data to understand user behavior and improve the app. Using Cloud SQL, Versa analyzes data from different sources such as [Google Analytics](https://marketingplatform.google.com/about/analytics/) (<https://marketingplatform.google.com/about/analytics/>) for marketing campaign data, [Firebase](https://firebase.google.com/) (<https://firebase.google.com/>) for user authentication, and AppsFlyer for mobile apps analytics.

“By running queries on Cloud SQL, we can pull out data points such as how customers interact with the app or how long it takes for users to complete a transaction. As a result, Versa can highlight features that customers like and resolve bottlenecks,” says Wong.

For example, Versa observed that users of a specific mobile phone model and OS version took longer to open an account. This data helped developers fix the bug and improve the onboarding process for a better customer experience.

To bring data to life, Versa uses Looker Studio (<https://marketingplatform.google.com/about/data-studio/>)

, Google's data visualization platform, that creates interactive dashboards and reports by connecting Cloud SQL and other data sources. With the help of Looker Studio, employees can easily make sense of data, such as a spike in customer acquisition linked to a marketing event.

"Looker Studio makes it easy for us to turn data into graphs without coding, so developers can focus on improving the platform instead of creating reports from scratch," says Wong. "From operations to customer service, everyone at Versa uses Looker Studio to understand how the company is performing over time and work together to achieve business goals."

Introducing a new way of managing your money.

No complications. No lock-in period. No hidden fees.
Meet Versa, a digital cash management platform that earns you interest that is on-par with Fixed Deposit (FD) and gives you the power to withdraw your funds anytime you wish, without penalties.



Managing security risks and regulatory compliance on BigQuery

As a Securities Commission Malaysia-regulated provider, Versa is required to monitor and flag money laundering, terrorism, and other illegal activities. With the help of BigQuery, Versa searches log event data to identify patterns of suspicious

transactions, such as rapid deposits and withdrawals in a short time.

According to Wong, “Analyzing log files in text format can be tedious and time-consuming. Instead of taking days, we can process massive amounts of data with BigQuery and generate insights in seconds.”

Each of Versa’s platform services generates its own set of logs. Versa can quickly query different log streams on BigQuery, without specific tags, for in-depth analysis. To date, Versa stores 140 GB of such log events in BigQuery, and that data is growing at one gigabyte per day.

Simplifying collaboration on Google Workspace

Google Workspace (<https://workspace.google.com/>) is a huge part of everyday life at Versa. Employees share and store files on Drive (<https://workspace.google.com/products/drive/>), create reports on Sheets (<https://workspace.google.com/products/sheets/>), and set up Meet (<https://workspace.google.com/products/meet/>) for sharing new ideas with colleagues and partners.

“Without Google Workspace, working from home would have been a nightmare, especially during the COVID-19 pandemic,” says Wong. “We’re able to work anywhere with secure access while legacy companies running on-premises servers and paper

processes can't adapt quickly to remote work because they are locked out of their offices."

In a growing organization, administrators often wonder, "How do I manage user accounts? How do I manage access controls when employees join and leave the company? How do I ensure that the right people get access to the back office and dashboards? Versa uses Google Cloud for single sign-on (SSO) to authenticate employees with their Gmail (<https://workspace.google.com/products/gmail/>) accounts and provide the right level of access, so they can work efficiently without compromising security.



Competing in the future of digital financial services

Versa's vision is to help Malaysians do more with their money. Following its successful launch in 2021, the company plans to introduce more features and enhancements into the application to provide more

opportunities to its customers to maximize their cash and savings potential.

“Every time we introduce a new service, we’re creating new tables in the databases with coding for API, integration work with the front end, and security testing to ensure that only developers with the correct credentials can gain API access,” says Wong. “Operating containers in GKE makes it easy for developers to build, scale, test, and deploy workloads independently without affecting other services.”

Wong adds, “There's a lot more for us to explore in Google Cloud offerings that we have not utilized at this point. I look forward to making the product better together with Google Cloud and MatrixC.”